8.1.5 Computing limits with equivalences

Fundamental limits

We have the following, all as $x \to 0$: *(these are the fundamental limits of Section 6.7, written with the symbol $\sim$)*

$$\begin{aligned} \sin x &\sim x \\ 1 - \cos x &\sim \frac{1}{2}x^2 \\ \ln(1+x) &\sim x \\ e^x - 1 &\sim x \\ (1+x)^\alpha - 1 &\sim \alpha x \end{aligned}$$

Proposition 8.3: Consider functions $f, \tilde{f}, g, \tilde{g} : \mathbb{R} \to \mathbb{R}$ such that $f \sim \tilde{f}$ and $g \sim \tilde{g}$ as $x \to x_0$. If $\lim_{x\to x_0} \tilde{f}(x)\tilde{g}(x)$ exists (finite or infinite), then so does $\lim_{x\to x_0} f(x)g(x)$, and

$$\lim_{x\to x_0} f(x)g(x) = \lim_{x\to x_0} \tilde{f}(x)\tilde{g}(x).$$

Similarly, if $\lim_{x\to x_0} \frac{\tilde{f}(x)}{\tilde{g}(x)}$ exists (finite or infinite), then so does $\lim_{x\to x_0} \frac{f(x)}{g(x)}$, and

$$\lim_{x\to x_0} \frac{f(x)}{g(x)} = \lim_{x\to x_0} \frac{\tilde{f}(x)}{\tilde{g}(x)}.$$

Proof. Since $\lim_{x\to x_0} \frac{g(x)}{\tilde{g}(x)} = 1 > 0$, we have $\frac{g(x)}{\tilde{g}(x)} > 0$, and in particular $g(x) \neq 0$, on a deleted neighborhood of x_0, so all the quotients below are defined.

Let us prove the first claim. We multiply and divide by the *tilde* functions as follows

$$\begin{aligned} \lim_{x\to x_0} (f(x)g(x)) &= \lim_{x\to x_0} \left(\frac{f(x)}{\tilde{f}(x)} \frac{g(x)}{\tilde{g}(x)} \tilde{f}(x)\tilde{g}(x) \right) \\ &= \left(\lim_{x\to x_0} \frac{f(x)}{\tilde{f}(x)} \right) \left(\lim_{x\to x_0} \frac{g(x)}{\tilde{g}(x)} \right) \left(\lim_{x\to x_0} \tilde{f}(x)\tilde{g}(x) \right) = \lim_{x\to x_0} \left(\tilde{f}(x)\tilde{g}(x) \right) \end{aligned}$$

where the equality going from the first line to the second uses the fact that it is not an indeterminate form. This proves the first claim. The second claim follows in the same way, writing $\frac{f}{g} = \frac{f}{\tilde{f}} \cdot \frac{\tilde{g}}{g} \cdot \frac{\tilde{f}}{\tilde{g}}$. □

Warning!

These rules do not apply to *sums* and *differences*. For instance, consider

$$\begin{aligned} f(x) &= x & \tilde{f}(x) &= x + 1 \\ g(x) &= x & \tilde{g}(x) &= x \end{aligned}$$

Then $f \sim \tilde{f}$ and $g \sim \tilde{g}$ as $x \to +\infty$. However,

$$\lim_{x\to+\infty} (f(x) - g(x)) = \lim_{x\to+\infty} 0 = 0 \neq 1 = \lim_{x\to+\infty} 1 = \lim_{x\to+\infty} (\tilde{f}(x) - \tilde{g}(x)).$$

Corollary 8.4: Consider functions $f, f_1, g, g_1 : \mathbb{R} \to \mathbb{R}$ and assume that $f_1 = o(f)$ and $g_1 = o(g)$ both as $x \to x_0$. Then

$$\lim_{x\to x_0} (f(x) \pm f_1(x))(g(x) \pm g_1(x)) = \lim_{x\to x_0} f(x)g(x)$$
$$\lim_{x\to x_0} \frac{f(x) \pm f_1(x)}{g(x) \pm g_1(x)} = \lim_{x\to x_0} \frac{f(x)}{g(x)}$$

Corollary 8.5: Consider functions $f, \tilde{f}, g, \tilde{g} : \mathbb{R} \to \mathbb{R}$ such that $f \sim \tilde{f}$ and $g \sim \tilde{g}$ as $x \to x_0$. Then

$$f = \mathcal{O}(g) \quad \Leftrightarrow \quad f = \mathcal{O}(\tilde{g}) \quad \Leftrightarrow \quad \tilde{f} = \mathcal{O}(g) \quad \Leftrightarrow \quad \tilde{f} = \mathcal{O}(\tilde{g})$$
$$f = o(g) \quad \Leftrightarrow \quad f = o(\tilde{g}) \quad \Leftrightarrow \quad \tilde{f} = o(g) \quad \Leftrightarrow \quad \tilde{f} = o(\tilde{g})$$

all as $x \to x_0$.

Example 8.3: Compute

$$\lim_{x\to 0} \frac{\sin 2x + x^3}{4x + 5\ln(1 + x^2)}.$$

1. First we simplify the numerator. We know that $\lim_{x\to 0} \frac{\sin x}{x} = 1$. This implies that $\sin 2x \sim 2x$ as $x \to 0$. Hence (using Corollary 8.5):

$$x^3 = o(2x) = o(\sin 2x), \quad x \to 0.$$

2. Now we turn to the denominator. We use the fact (see the Fundamental limits box above) that $\ln(1 + x) \sim x$ as $x \to 0$. Hence, $\ln(1 + x^2) \sim x^2$ as $x \to 0$. Hence:

$$5\ln(1 + x^2) \sim 5x^2 = o(4x), \quad x \to 0.$$

3. So we have:

$$\lim_{x\to 0} \frac{\sin 2x + x^3}{4x + 5\ln(1 + x^2)} = \lim_{x\to 0} \frac{\sin 2x}{4x} = \frac{1}{2} \underbrace{\lim_{x\to 0} \frac{\sin 2x}{2x}}_{=1} = \frac{1}{2}.$$

Important takeaway

When we want to study the limit of a complicated expression, we need to understand the asymptotic behavior of all the terms that it includes, and try to convert them to monomials.

8.2 Infinitesimal and infinite functions

As we have seen in the previous section, we are interested in the asymptotic study of the behavior of functions as we either approach their zeros (i.e. points where the function vanishes) or where they tend to $\pm\infty$. In what follows we let $x_0 \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$.

If $\lim_{x\to x_0} f(x) = 0$ then f is said to be **infinitesimal** at x_0.
If $\lim_{x\to x_0} f(x) = \pm\infty$ then f is said to be **infinite** at x_0.

1. If f and g are two *infinitesimal* functions at x_0, then:
   - if $f = o(g)$ at x_0 then f is said to be *infinitesimal of a higher order* than g. Sometimes we write:
     $$|f| \ll |g| \ll 1$$
     to signify that $f = o(g)$ and $g = o(1)$.
   - if $f \asymp g$ at x_0 then f and g are said to be *infinitesimal of the same order*.

2. If f and g are two *infinite* functions at x_0, then:
   - if $f = o(g)$ at x_0 then f is said to be *infinite of a lower order* than g. Sometimes we write:
     $$1 \ll |f| \ll |g|$$
     to signify that $f = o(g)$ and $\lim_{x\to x_0} |f| = +\infty$.
   - if $f \asymp g$ at x_0 then f and g are said to be *infinite of the same order*.

Ordering of important infinite functions

The following functions are ordered in terms of their infinite order as $x \to +\infty$:
(we will prove this in Chapter 10, using de l'Hôpital's Theorem)

$$\log_a x \qquad x^s \qquad b^x$$

for any $a > 1,\ s > 0,\ b > 1$.

This means that:

$$\left.\begin{aligned} \log_a x &= o(x^s) \\ x^s &= o(b^x) \end{aligned}\right\} \quad \text{as } x \to +\infty$$

Ordering of important infinite sequences

The following sequences are ordered in terms of their infinite order as $n \to \infty$:
(see the proof below)

$$\log_a n \qquad n^s \qquad b^n \qquad n! \qquad n^n$$

for any $a > 1,\ s > 0,\ b > 1$.

This means that:

$$\left.\begin{aligned} \log_a n &= o(n^s) \\ n^s &= o(b^n) \\ b^n &= o(n!) \\ n! &= o(n^n) \end{aligned}\right\} \quad \text{as } n \to \infty$$

Indeed, the first relation follows from the corresponding relation for functions (take $x = n$). The other three follow from the Ratio Test (Theorem 6.16): for each of the following positive sequences, the ratio of consecutive terms tends to a limit $q < 1$, so the sequence tends to 0:

$$\begin{aligned} \frac{n^s}{b^n}: &\qquad \frac{(n+1)^s/b^{n+1}}{n^s/b^n} = \left(1+\frac{1}{n}\right)^s \frac{1}{b} \to \frac{1}{b} < 1, \\ \frac{b^n}{n!}: &\qquad \frac{b^{n+1}/(n+1)!}{b^n/n!} = \frac{b}{n+1} \to 0, \\ \frac{n!}{n^n}: &\qquad \frac{(n+1)!/(n+1)^{n+1}}{n!/n^n} = \left(\frac{n}{n+1}\right)^n = \frac{1}{\left(1+\frac{1}{n}\right)^n} \to \frac{1}{e} < 1. \end{aligned}$$

Stirling formula

There is a precise formula for the relationship between $n!$ and n^n, known as **Stirling's formula**, which provides a way to approximate $n!$:

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n, \qquad n \to \infty.$$

This is useful in many applications, including in statistics, where factorials appear in the binomial formula.

We don't prove it here. It is remarkable that the constants π and e both appear in a formula about $n!$, which involves only multiplication of integers.

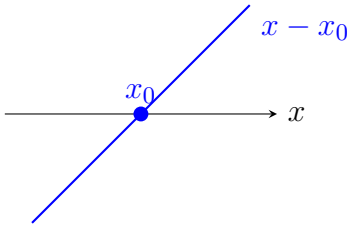
8.3 Order and principal part of infinitesimals and infinites

To understand the behavior of an infinite or infinitesimal function, it is simplest to compare it to powers of x.

Infinitesimal functions at a finite point x_0

Suppose that $x_0 \in \mathbb{R}$, and that $\lim_{x\to x_0} f(x) = 0$. Then

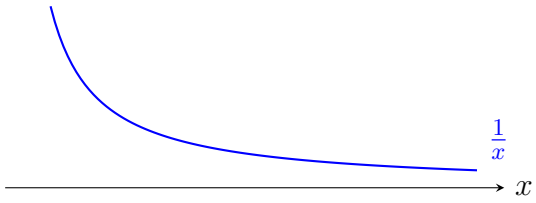
$$\textit{we want to compare } f \textit{ to powers of } \varphi(x) = x - x_0$$



Infinitesimal functions at an infinite point x_0

Suppose that $x_0 \in \{\pm\infty\}$, and that $\lim_{x\to x_0} f(x) = 0$. Then

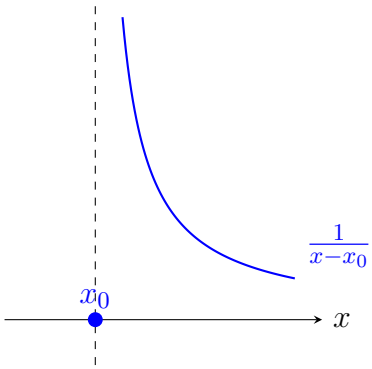
we want to compare f *to powers of* $\varphi(x) = \dfrac{1}{x}$



Infinite functions at a finite point x_0

Suppose that $x_0 \in \mathbb{R}$, and that $\lim_{x\to x_0} f(x) = \pm\infty$. Then

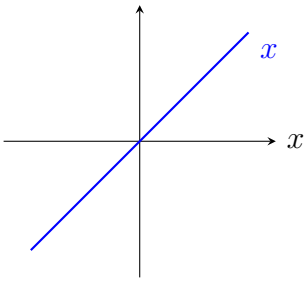
we want to compare f *to powers of* $\varphi(x) = \dfrac{1}{x - x_0}$



Infinite functions at an infinite point x_0

Suppose that $x_0 \in \{\pm\infty\}$, and that $\lim_{x\to x_0} f(x) = \pm\infty$. Then

we want to compare f *to powers of* $\varphi(x) = x$



If α is not an integer, φ^α is only defined where $\varphi > 0$. Since $x - x_0$ and $\frac{1}{x-x_0}$ are negative for $x < x_0$, and x and $\frac{1}{x}$ are negative as $x \to -\infty$, in these cases we use $|\varphi|$ instead of φ (for example $|x - x_0|^\alpha$).

Order of an infinitesimal/infinite function

Let f be infinitesimal or infinite at x_0. If there exists $\alpha > 0$ such that

$$f \asymp \varphi^\alpha, \quad x \to x_0$$

then we say that α is the **order** of f at x_0 with respect to φ.

Observe that if f has an order it is *unique*. Furthermore, by definition of the symbol $\asymp$, $f \asymp \varphi^\alpha$, $x \to x_0$, means that

$$\lim_{x \to x_0} \frac{f(x)}{\varphi^\alpha(x)} = \ell \in \mathbb{R} \setminus \{0\}.$$

This can also be written as:

$$f \sim \ell\varphi^\alpha, \quad x \to x_0$$

which can be rewritten as

$$f = \ell\varphi^\alpha + o(\varphi^\alpha), \quad x \to x_0.$$

Principal part of an infinitesimal/infinite function

If $f = \ell\varphi^\alpha + o(\varphi^\alpha)$ as $x \to x_0$ then

$$p(x) = \ell\varphi^\alpha(x)$$

is called the **principal part** of f at x_0 with respect to φ.

Examples

Example 8.4: Consider the function

$$f(x) = \sin x - \tan x, \qquad \text{near } x_0 = 0.$$

Since $\sin 0 = \tan 0 = 0$, f is infinitesimal at $x = 0$. We can write:

$$\begin{aligned} \sin x - \tan x &= \sin x - \frac{\sin x}{\cos x} \\ &= \frac{\sin x \cdot (\cos x - 1)}{\cos x} \\ &\sim \frac{x \cdot (-\frac{1}{2}x^2)}{1} \\ &= -\frac{1}{2}x^3, \qquad x \to 0. \end{aligned}$$